Sepehr Tabrizchi

Ph.D. in Electrical & Computer Engineering | **Hardware Architecture & RTL**
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Authorized to work in the U.S. (no sponsorship required)

Summary

Hardware engineer and computer architect (Ph.D., Electrical and Computer Engineering; M.Sc., Computer Architecture) with 8+ years spanning **microarchitecture**, **Verilog/SystemVerilog RTL**, **FPGA prototyping**, **mixed-signal circuits**, and **C++/Python performance modeling**, with a focus on energy-efficient AI accelerators and memory systems. Built a RISC-V processor with custom ML instructions whose single-cycle and 5-stage RV32IM cores pass all 48 official tests, designed in-memory and in-sensor neural-network accelerators, and built simulators that guide architecture decisions within ∼5% of SPICE. Works across the hardware/software boundary, from circuits and RTL to firmware and ML models. 27 journal and 38 conference papers (DAC, DATE, ISLPED, FCCM, IEEE TC, TCAD), h-index 17, and a published U.S. patent application with three more pending.

Skills

- **RTL & FPGA**: Verilog, SystemVerilog, VHDL; pipelined datapaths and FSM control; self-checking SystemVerilog testbenches; ModelSim/QuestaSim; Synopsys Design Compiler; Xilinx Vivado (Artix-7, Zynq-7000), ILA; FPGA bring-up of streaming and processor designs.
- **Architecture**: AI/ML accelerators (MAC and convolution datapaths, quantization), memory hierarchy and memory-controller scheduling, DRAM in-memory computing, RISC-V and ISA extensions for ML kernels, intermittent computing, PPA trade-offs.
- **Modeling & Software**: Cycle- and instruction-accurate simulators in C++ and Python, validated against SPICE and hardware; Python, C/C++, Bash, Linux, Git; PyTorch, TensorFlow; int8/binary quantization; LLM-assisted code optimization (CHAI).
- **Circuits**: Cadence Virtuoso, HSPICE, DRC/LVS, Monte Carlo (45 nm); ADCs, sense amplifiers, comparators, readout circuits.
- **Embedded & Lab**: MCU firmware in C (Apollo4, ESP32, STM32); custom PCB design (Altium); oscilloscopes and logic analyzers.

Experience

- **Postdoctoral Researcher** — 2025 – Present
  University of Illinois Chicago — *Hardware accelerators, mixed-signal sensing, and system modeling*
  - **Architecture & Prototype**: Define a per-pixel dynamic frame-rate image sensor co-designed with its YOLO12n detector, prototyped on an Apollo4 Plus MCU driving a 32×32 photodiode array, trading data volume against detection accuracy.
  - **Mixed-Signal Design**: Pre-ADC analog convolution in the sensor front end for compressed-domain eye parsing (with UT Austin); HSPICE Monte Carlo mismatch and offset analysis of comparator and readout circuits (45 nm) to confirm operation under variation.
  - **HW/SW Co-design**: Built CHAI, an LLM-driven approximate-computing framework; with -O3, it cut execution time 71.4% and energy 71.1% on ESP32 hardware, with an LLM choosing and applying code transformations automatically.
  - **Modeling & Leadership**: Build performance and energy models of ML accelerators and energy-harvesting systems; mentor six students; write research proposals to Sony's Research Award Program (per-pixel frame-rate image sensors), NSF, and DARPA.
- **Graduate Research Assistant** — 2022 – 2025
  University of Illinois Chicago / University of Nebraska-Lincoln — *Accelerator microarchitecture, memory systems, and modeling*
  - **Accelerator Design**: Led AppCiP (IEEE JETCAS 2023), an approximate convolution-in-pixel design reaching ∼4.12 TOp/s/W at 3000 fps and the basis of a published U.S. patent application; also TizBin, Ocellus (ISLPED 2023 Best Paper Nominee), and APRIS.
  - **Datapaths**: Designed ResISC (DAC 2025), a carry-free RNS datapath integrated with the sensor (94.63% CIFAR-10 vs. 95.0% floating point).
  - **Memory Systems**: Co-developed PIMA-SIM (C/C++), a cycle-accurate simulator of memory-controller scheduling and bank organization across SRAM and NVMs; designed dual-mode DRAM in-memory computing and row-hammer mitigation.
  - **Modeling Tools**: Built PINSim (IEEE CAL 2025) and PiPSim (IEEE TCAD 2024), PPA simulators within ∼5% of HSPICE and over 25,000× faster.
  - **Low-Power Control**: Architected BISen (IEEE TC 2026), an energy-aware FSM for battery-free MCUs, cutting NVM reads/writes up to 86.4% and raising throughput 1.3–1.4× (open source as BISen-IC); also DiAC (DATE 2024) power-state control.
  - **RTL & Synthesis**: Wrote Verilog for accelerator control and peripheral logic; synthesized with Synopsys DC for area/power estimates.
- **Embedded System Programmer** — 2019 – 2021
  Hooshmand Rayaneh — *Industry: embedded hardware and software*
  - **Product Ownership**: Owned a closed-loop thermal management product end to end: custom Altium PCB, MCU firmware in C, and a C# host app.
- **Earlier Roles**: Software Developer, Makan System (2020; SAP2000-based structural-analysis platform); C++ Programmer, Sinc 3D (2016–2018; real-time FFT audio engine); C++/VHDL Programmer, Makan System (2012–2014; custom router); JavaScript Programmer, Mika System (2010–2012; PhoneGap game engine); Research Assistant and Instructor, Shahid Beheshti University (2017–2021; 13 journal papers; teaching).

Selected Hardware Projects

- **RISC-V Processor with Custom ML Instructions (SystemVerilog, PYNQ-Z2)**: 5-stage RV32IM core with custom int8 ML instructions: 48/48 riscv-tests, 65 MHz on the board, 4.37× faster quantized MNIST inference. GitHub
- **AES-128 Streaming Pipeline (Verilog, Nexys Video / Artix-7)**: Fully pipelined encryption/decryption of live audio, optimized for throughput and latency; verified against known-answer tests with SystemVerilog testbenches and run on the board with live audio.
- **8-bit Full-Custom ALU (Virtuoso, HSPICE)**: Schematic to DRC/LVS-clean layout; timing and power optimized through transistor sizing.

Education

- **Ph.D., Electrical and Computer Engineering**, University of Illinois Chicago (GPA 4.0) — 2022 – 2025
- **M.Sc., Computer Architecture**, IAU Science and Research Branch, Tehran — 2014 – 2017
- **B.Sc., Computer Engineering (Hardware)**, IAU Najafabad, Isfahan — 2008 – 2014

Publications, Patents & Awards

- **Publications**: 27 journal and 38 conference papers, h-index 17. Selected: *BISen* (IEEE TC 2026), *ResISC* (DAC 2025), *PINSim* (IEEE CAL 2025), *PiPSim* (IEEE TCAD 2024), *APRIS* (IEEE TETC 2024), *DiAC* (DATE 2024), *AppCiP* (IEEE JETCAS 2023), *Ocellus* (ISLPED 2023).
- **Patent**: U.S. Patent Application US 2025/0218169 A1, *Neural Network Acceleration of Image Processing* (co-inventor); three more applications pending.
- **Awards & Service**: Best Paper Nominee (IGSC 2026, ISLPED 2023), DAC Young Fellow (2023). DATE 2027 and FCCM 2025 program committees; reviewer for ICCAD, ISLPED, GLSVLSI, IEEE TCAS-I, IEEE JETCAS; UNL Outstanding Graduate Student Research Award (2023–24).